

Assignment 1 (14/07/2026)

1. Make a triangle signal at 200 Hz and make a wave (duration: 0.5 sec) that samples it at 10000 frames per second. Plot the signal for 3 periods and show the spectrum along with the harmonics. Listen to the wave.
2. Make a square signal at 100 Hz and make a wave (duration: 0.5 sec) that samples it at 10000 frames per second. Plot the signal for 3 periods and show the spectrum along with the harmonics. Listen to the wave.
3. Make a triangle signal at 1100 Hz and make a wave that samples it at 10000 frames per second. Plot the signal for 3 periods and show the spectrum along with the harmonics. If you plot the spectrum, you can see that most of the harmonics are aliased. When you listen to the wave, can you hear the aliased harmonics?
4. Make a square signal at 1100 Hz and make a wave that samples it at 10000 frames per second. Plot the signal for 3 periods and show the spectrum along with the harmonics. If you plot the spectrum, you can see that most of the harmonics are aliased. When you listen to the wave, can you hear the aliased harmonics?
5. Make a cosine signal at 4500 Hz and 5500 Hz that samples it at 10000 frames per second. Plot the signal for 5 periods and show the spectrum. What do you see in these two spectrums? Do you see any problem?

Assignment Submission Due: 18/07/2026